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PTH STA

# PlasticO:
# An ontology for plastic and plastic pollution

Giorgio A. Ubbiali and Ludger Jansen

PTH Brixen College

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# Plastics and Plastic Pollution

- Plastics are largely used and beneficial in disparate fields, including constructions, agriculture, and daily use.

- Several pollution concerns.
  - Alteration of soil biogeochemical element cycles.
  - Greenhouse gas emissions.
  - Water contamination and trophic transfer of microplastics.
  - Microplastic infiltration of organisms' bodies.

  *Mutual interconnections with climate change.*

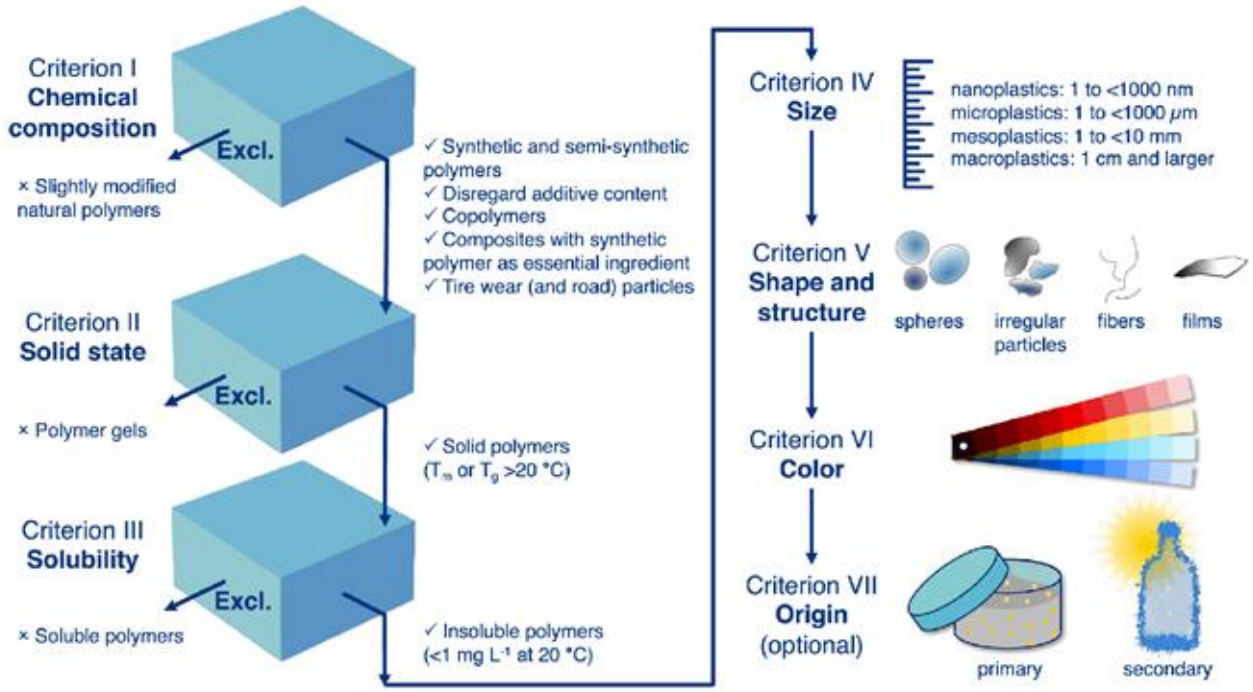
*(Eerkes-Medrano & Thompson, 2018 ; FAO, 2021; Filipe et al., 2023; Hartmann et al., 2019; Kelly et al., 2025; Lakhiam et al., 2024; Masterton et al., 2023; Ojukwu et al., 2024; Rochman et al., 2019 Yu et al., 2022; Yuan et al., 2022)*





# Plastics and Climate Change Semantics

- Cross-domain studies.
  - Vague terminologies.
  - Disparate standards of reference.
  - Various systems of classification.




*An account of plastic debris. From Hartman et al. (2019).*

*An attractive opportunity: **Ontologies**.*

*But to date, no ontology addresses plastic pollution and its interrelations with climate change.*

# The ClimOO Ontology Suite

- A reference ontology suite for representing plastics, (micro-)plastic pollution, climate change, and their interrelations.
- Use case: agrifood systems; South Tyrolean (Italy) apple production.

- Ontology Repo: https://w3id.org/climoo/repo
- ClimOO Project: https://pthsta.it/climoo
- English + Web Ontology Language (OWL).



# The ClimOO Suite Architecture

- A modular structure:
  - Plastics Ontology (PlasticO),
  - ClimOO-SDC2,
  - Plastics Interconnecting with Climate Change Ontology (PICCO),
  - PICCOmela.
- Extending Basic Formal Ontology (BFO) and Sustainability Core Ontology (SCO).

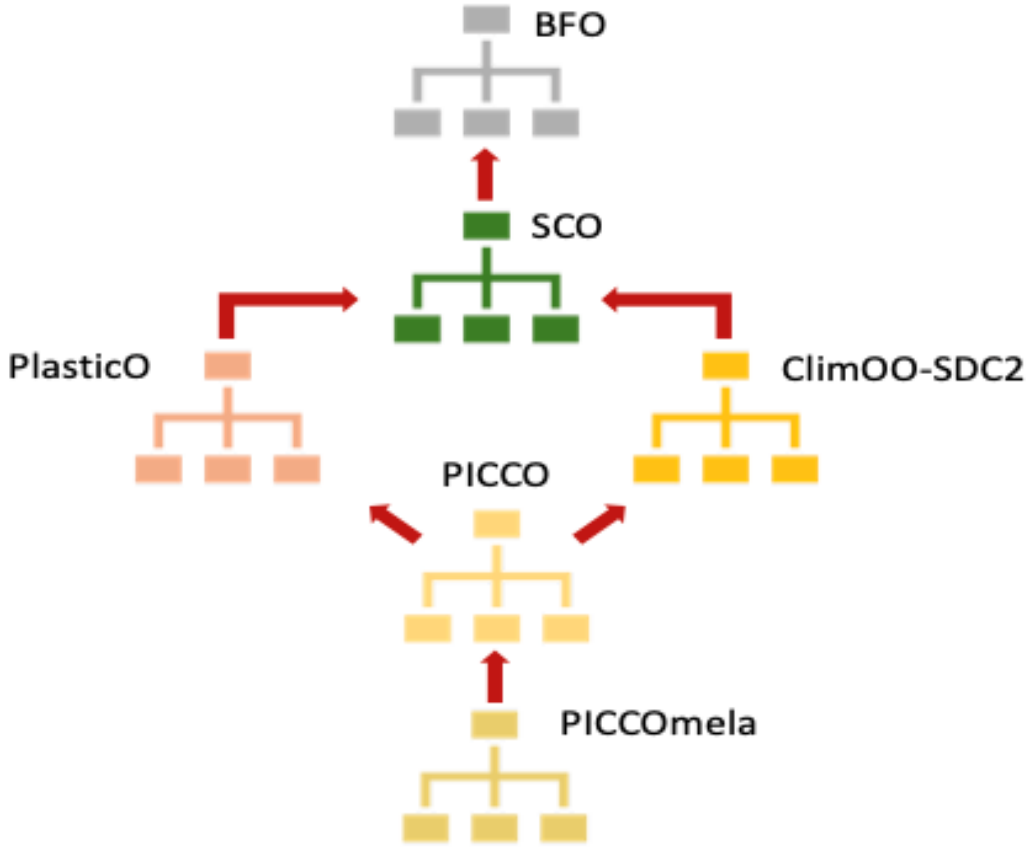



*Fig. The ClimOO Ontology suite structure.*

# The ClimOO Methodology

- Sustainable Meat Systems Ontology (SuMSO) methodology (Ubbiali et al., 2026).
  - Based on existing standards and methodologies such as METHONTOLOGY and the OTK methodology.

- Assessment of the relevant literature and major international standards.
- Stakeholder engagement.
- Reasoner validation and Competency Questions (CQs).
- Imports from existing resources, including EnvO, ChEBI, and PMDco.
- OBO Foundry principles.

*(Fernández-López et al., 1997; Staab et al., 2001; Ubbiali et al., 2026 )*





# The PlasticO Module

- Domain of reference: plastics and plastic life cycles, plastic debris, and plastic pollution processes.
- Transdisciplinary approach.

- Extending SCO and BFO and importing from 14 ontologies, mainly from the OBO Foundry and BFO environments.
- 463 classes, 273 object properties, and 8 individuals.
- https://www.w3id.org/climoo/climoo-plastico

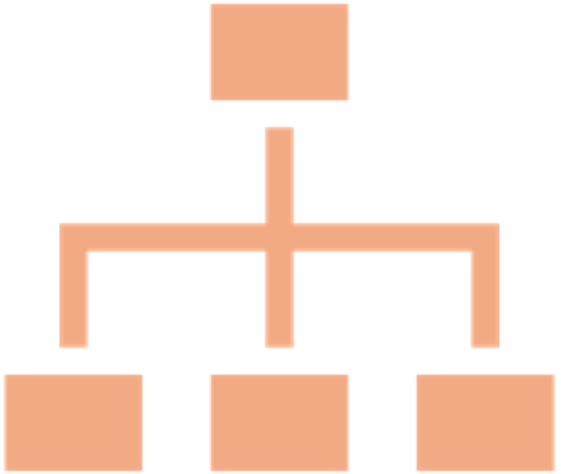
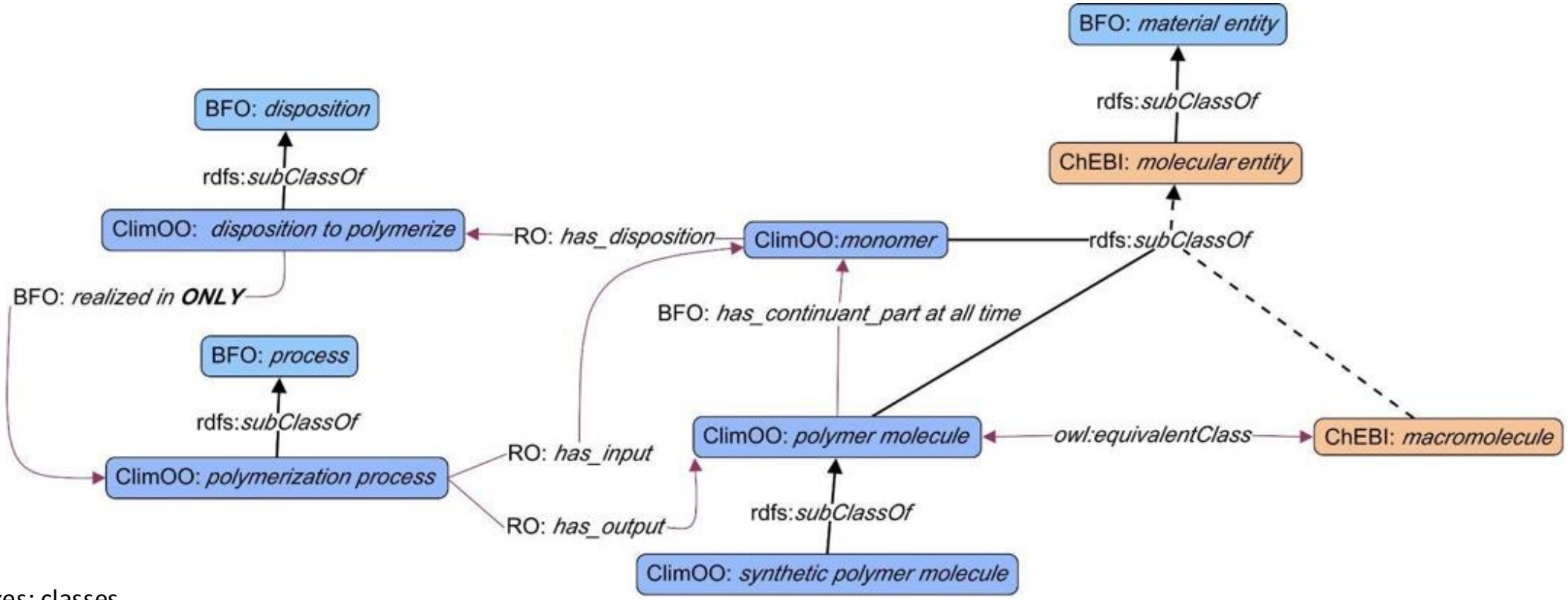


*Fig. The PlasticO module.*

# Polymers Molecules




Boxes: classes
**Bold arrows**: indirect is_a
Dashed arrows: indirect is_a
Red arrows: other relations

*Fig. Polymers and monomers. from Ubbiali and jansen (2026a).*

# Plastics and Pieces of Plastics

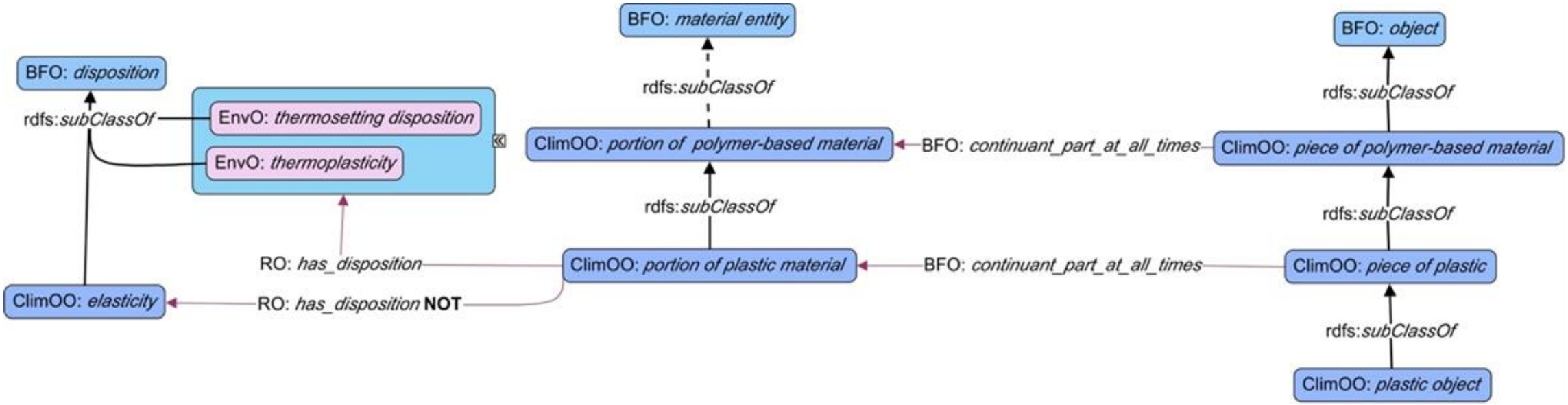



*Fig. Portions and pieces of plastics. from Ubbiali and jansen (2026a).*

# Plastic Life Cycle

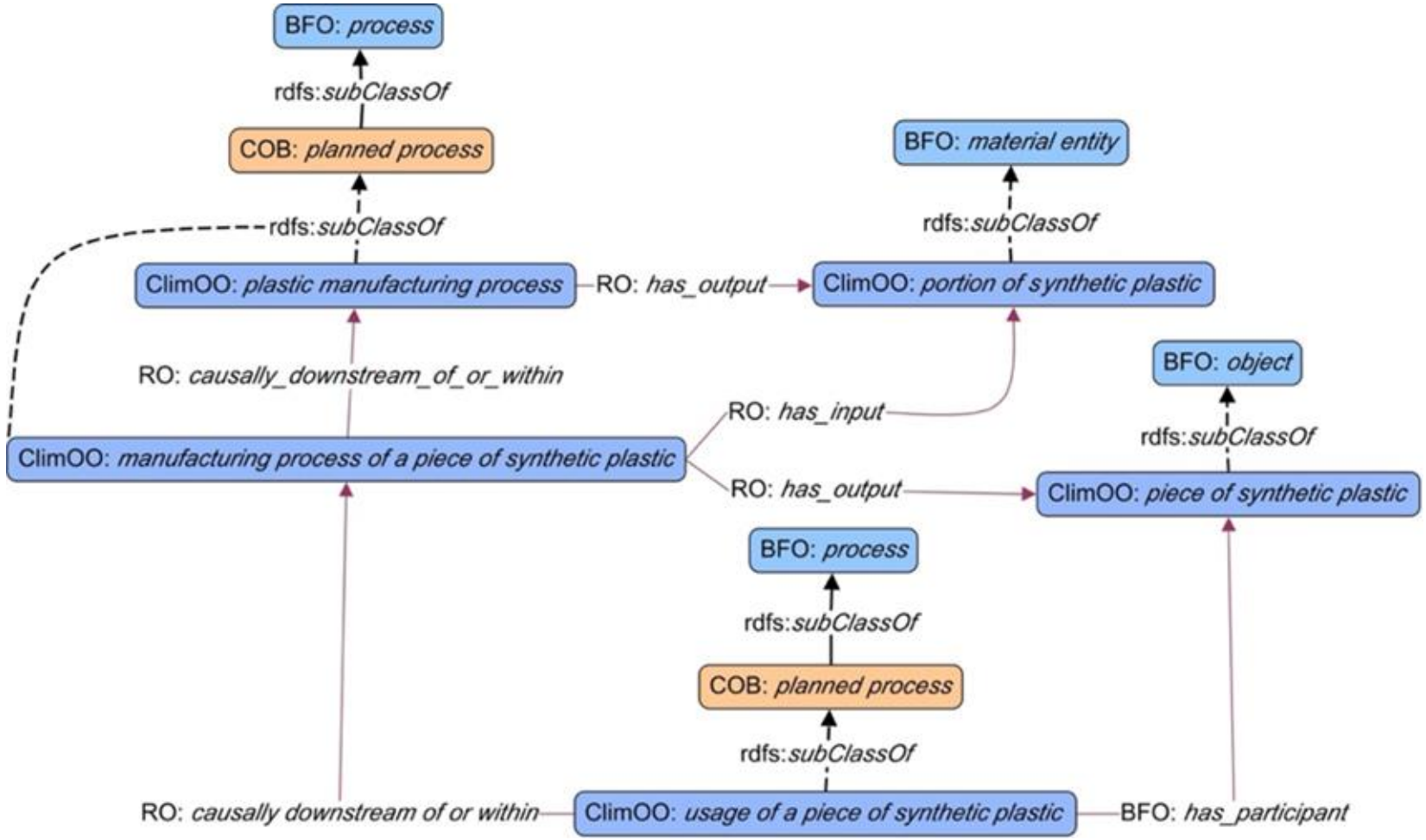



*Fig. Plastic life cycle. from Ubbiali and jansen (2026b).*

# Plastic Debris

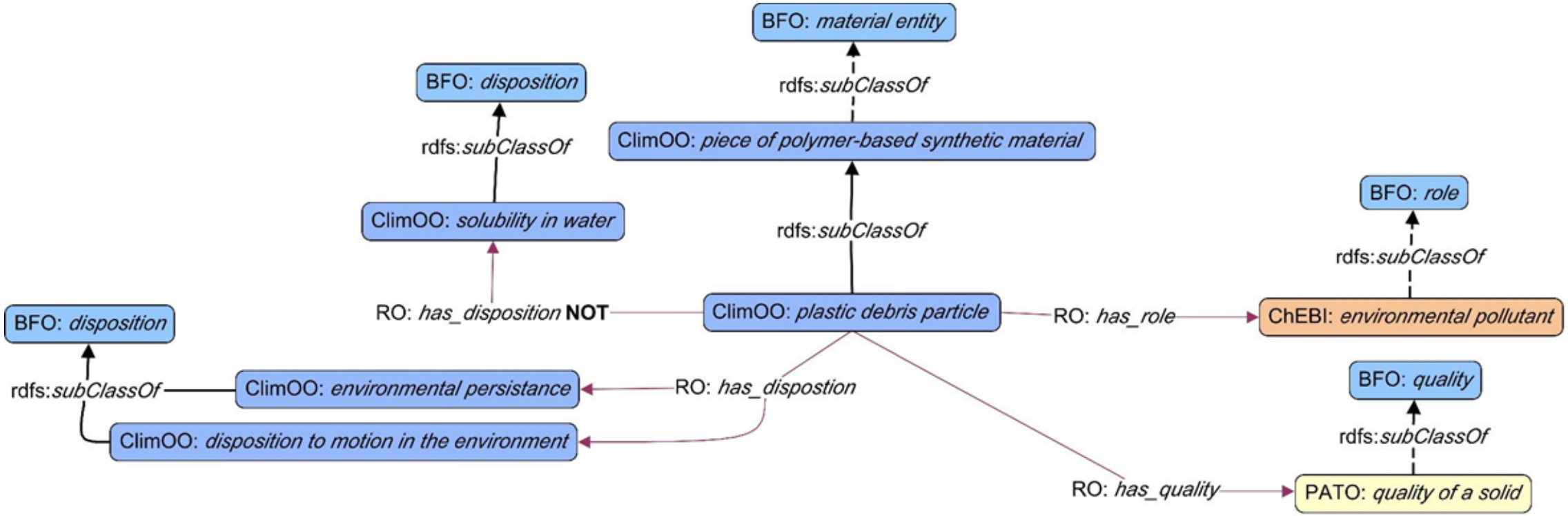



*Fig. Plastic debris particles. from Ubbiali and jansen (2026a).*

# Plastic Pollution

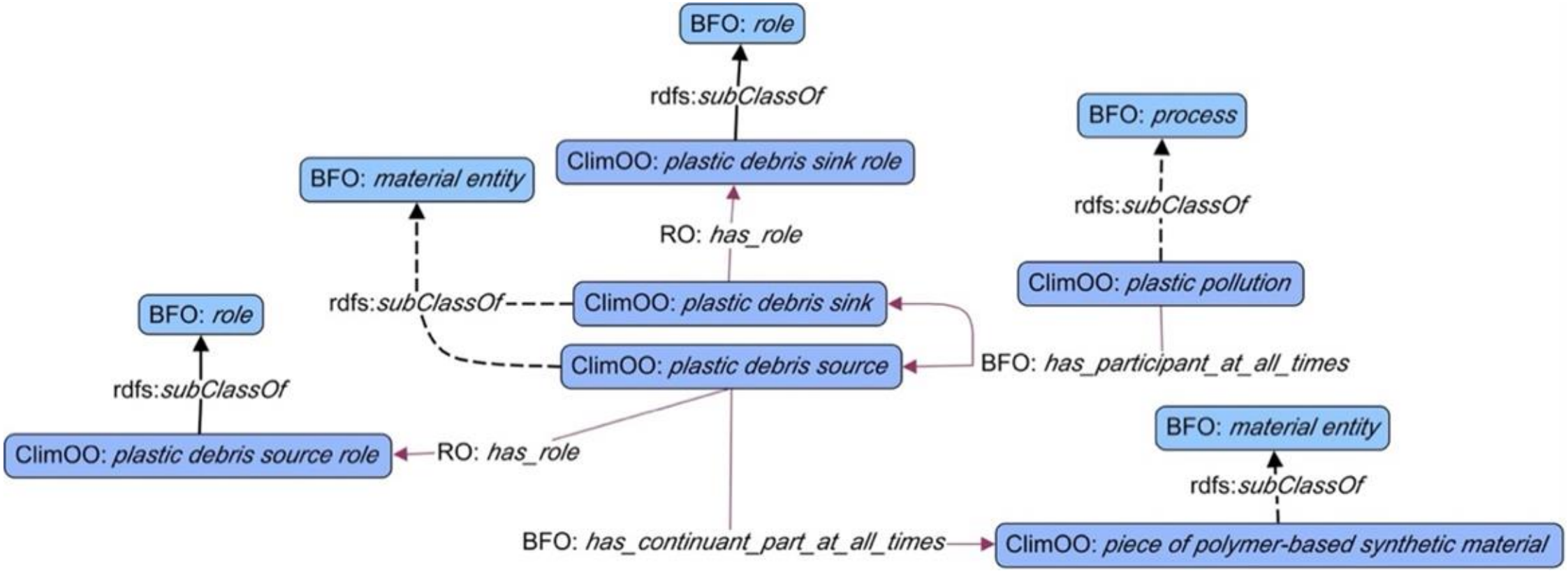



*Fig. Plastic pollution. from Ubbiali and jansen (2026b).*

# Conclusions and Ongoing Works

- The design of an ontology suite for plastic pollution and climate change: ClimOO.
- Ongoing works:
  - Construction of the PICCO and PICCOmela modules.
  - SPARQL queries for CQs testing.
  - Stakeholder approval.
  - Use case datasets for ontology implementation.

**COSMOS**: A Community of Common Ontologies, Semantics, and MOdels for Sustainability.





# References and Slide Deck



https://github.com/ClimOO/ClimOO-Ontology/tree/main/ClimOO%20material/materials%20from%20ClimOO%20presentations/5th%20Ontologies4Chem%20Workshop




# Thank You!!

Giorgio.Ubbiali@pthsta.it